Supplementary Information

Three-dimensional random-access confocal microscopy with 3D remote focusing system

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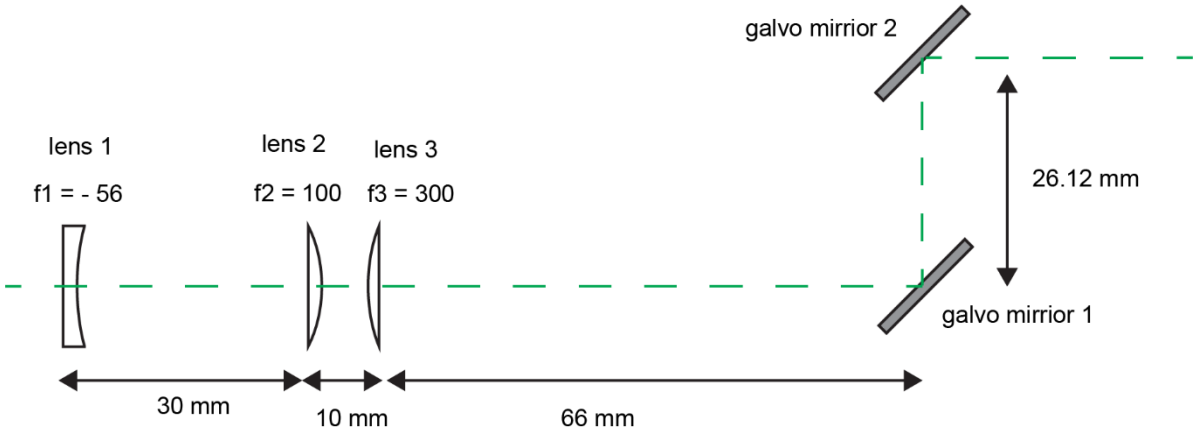


Figure S1. Detailed schematic of remote focusing system.

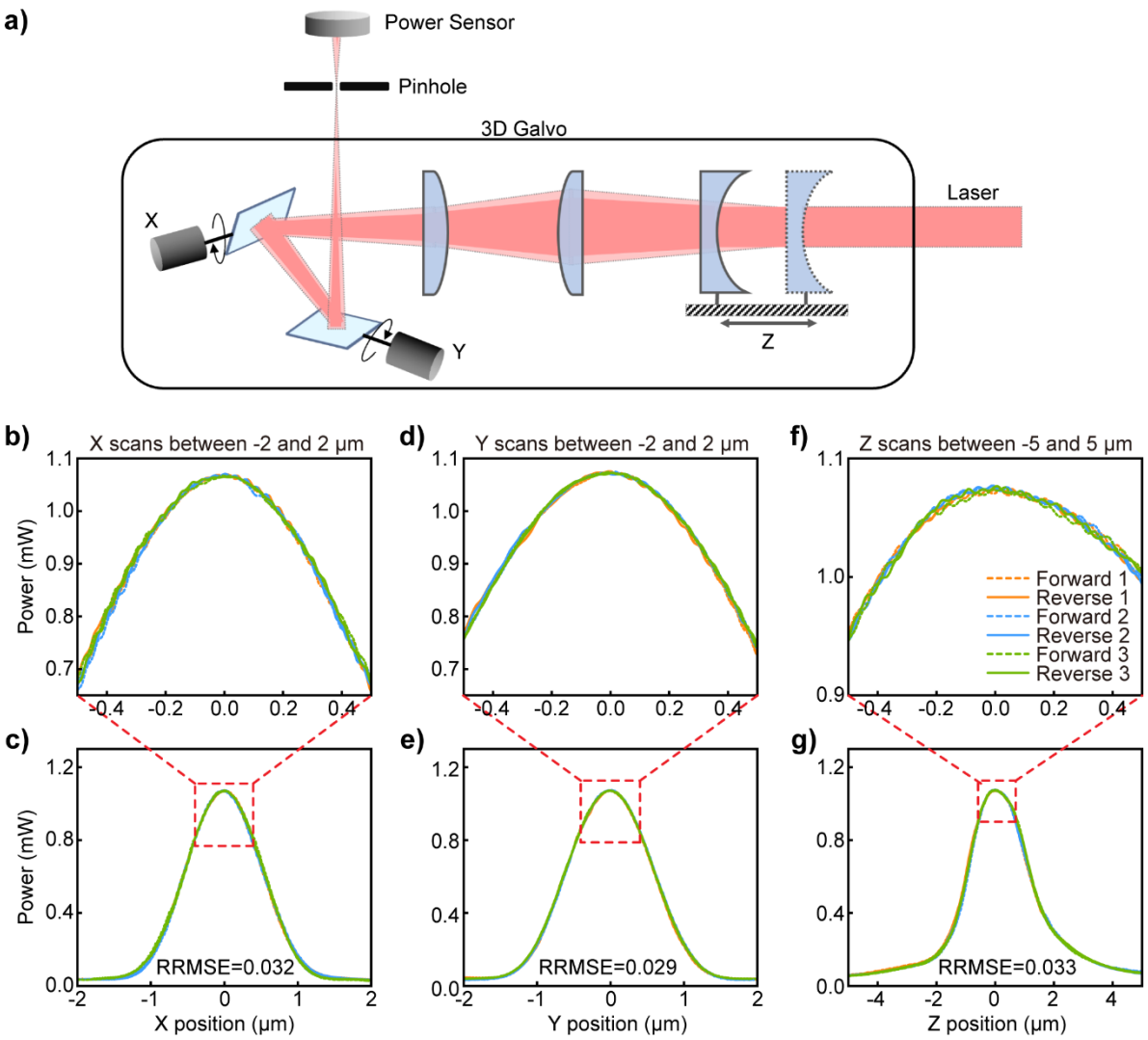


Figure S2. Bidirectional repeatability measurement of the 3D dynamic focusing system via power measurement after a pinhole. (a) Schematic of setup for testing the bidirectional repeatability of the 3D dynamic focusing system. A 100 μm pinhole was placed near the focal plane of 3D dynamic focusing system and a power meter was used to detect the light intensity after the pinhole. **(b-g)** The laser power measurement results in x **(b-c)**, y **(d-e)**, and z **(f-g)**, respectively. Three measurements were conducted, with forward scans represented by orange, blue, and green dashed lines, and reverse scans by solid lines. **(c, e, g)** Scanning results along the x, y, and z axes. The scanning range with ranges for x and y is -2 μm to 2 μm. The scanning range with ranges for z is -5 μm to 5 μm. **(b, d, f)** Enlarged views of the red dashed boxes in **(c, e, g)**, respectively. The relative root mean square error (RRMSE) values were calculated for characterizing the repeatbility. The RRMSE values for the x, y, and z axes were

0.032, 0.029, and 0.033, respectively, indicating good bidirectional repeatability across all axes.

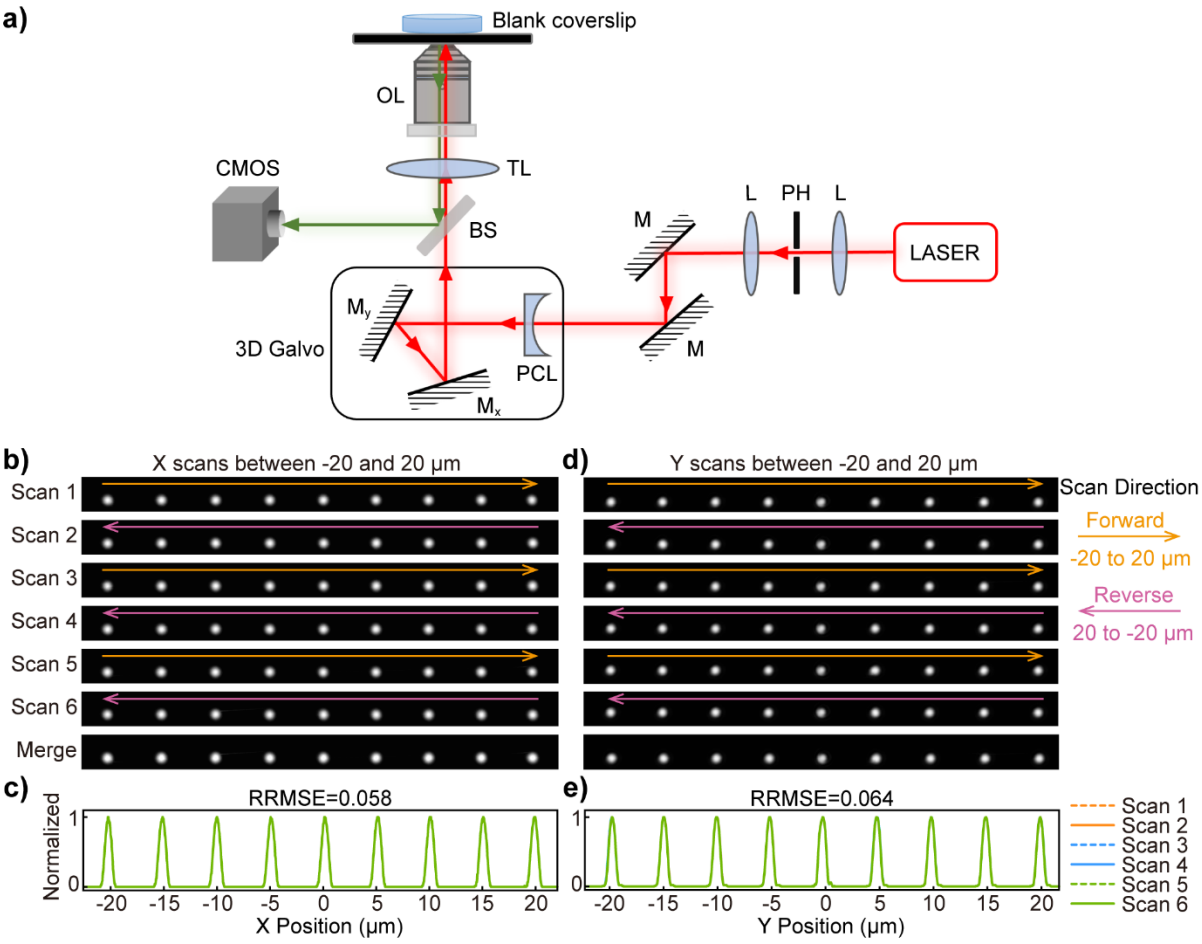


Figure S3. Bidirectional repeatability measurement of the 3D dynamic focusing system via scanning images. (a) Schematic of setup for testing the bidirectional repeatability of the 3D dynamic focusing system. The excitation light is focused on a blank coverslip and imaged on a CMOS camera. As the system scans in the x and y directions, the CMOS captures laser spot images at 5 μm intervals. **(b, d)** Scanning images in the range between -20 and 20 μm in x and y direction, respectively. Three sets of scanning results are shown: Scan 1 and Scan 2 represent the first set, Scan 3 and Scan 4 the second set, and Scan 5 and Scan 6 the third set. Each set shows both forward (orange arrow) and reverse (pink arrow) scan results. The bottom row presents the merged images from all six scans. **(c, e)** Overlayed normalized intensity profiles along the scanning direction. The RRMSE values for the x and y axes are 0.058 and 0.064, respectively.

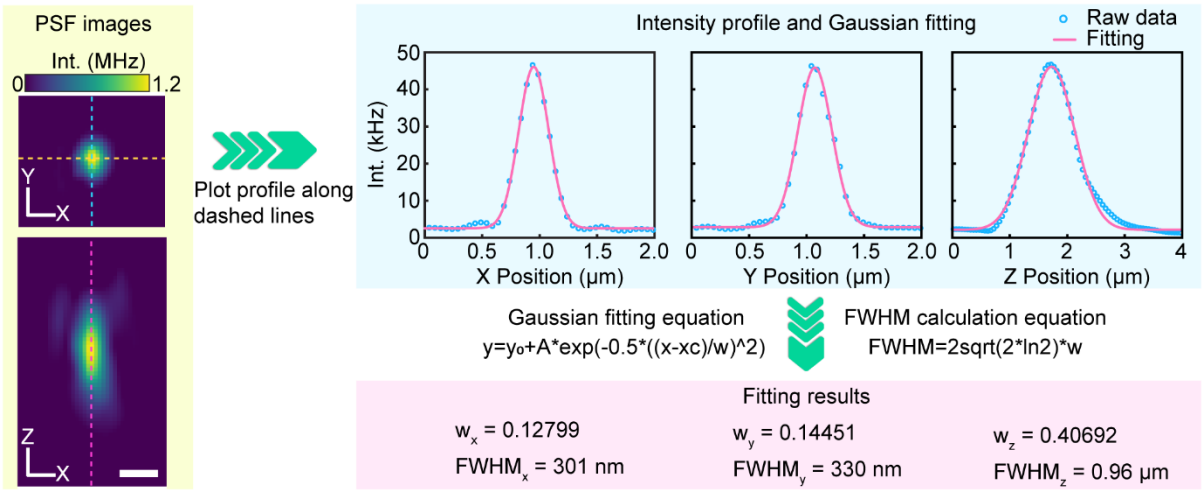


Figure S4. Resolution characterization with fluorescent sphere image. The intensity profiles along the x, y, and z directions are extracted and Gaussian fitted. The resolution is calculated by the full widths at half maximum (FWHM) as noted in the figure.

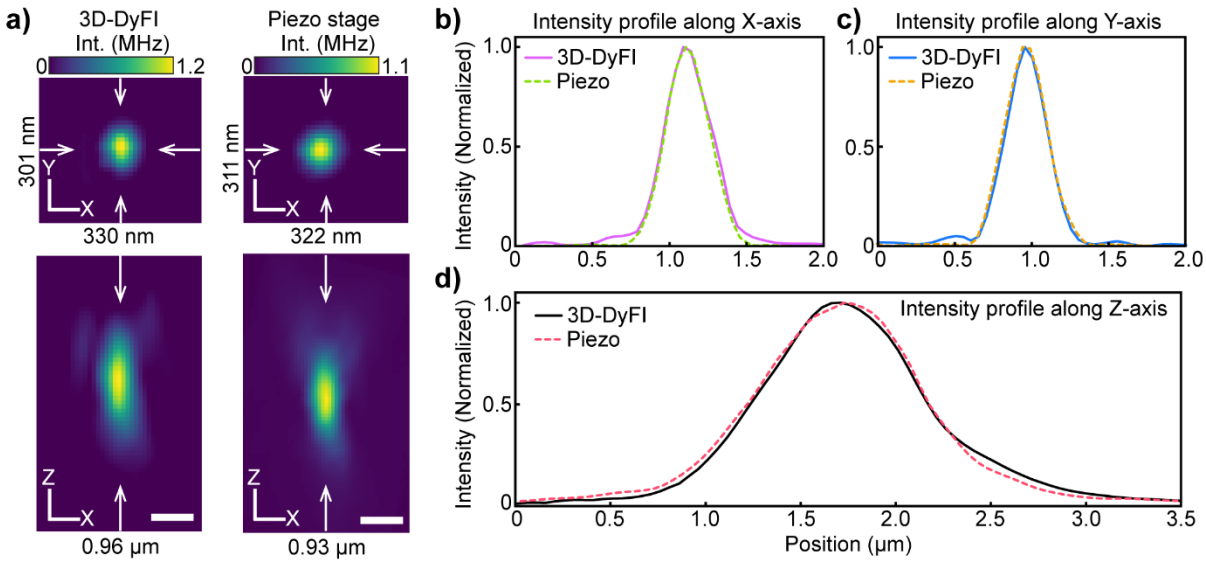


Figure S5. Comparison of the PSF between 3D-DyFI and the piezo-stage based confocal microscopy. **(a)** The scanning images of 50 nm fluorescent beads with 3D-DyFI and the piezo-stage based confocal microscopy. Upper panel: x-y plane section images; lower plane: x-z plane section images. The color indicates fluorescence intensity. **(b-d)** The comparative analysis of sectional intensity profiles of PSF images in **(a)** along x **(b)**, y **(c)**, and z **(d)** direction. Scale bar: 500 nm.

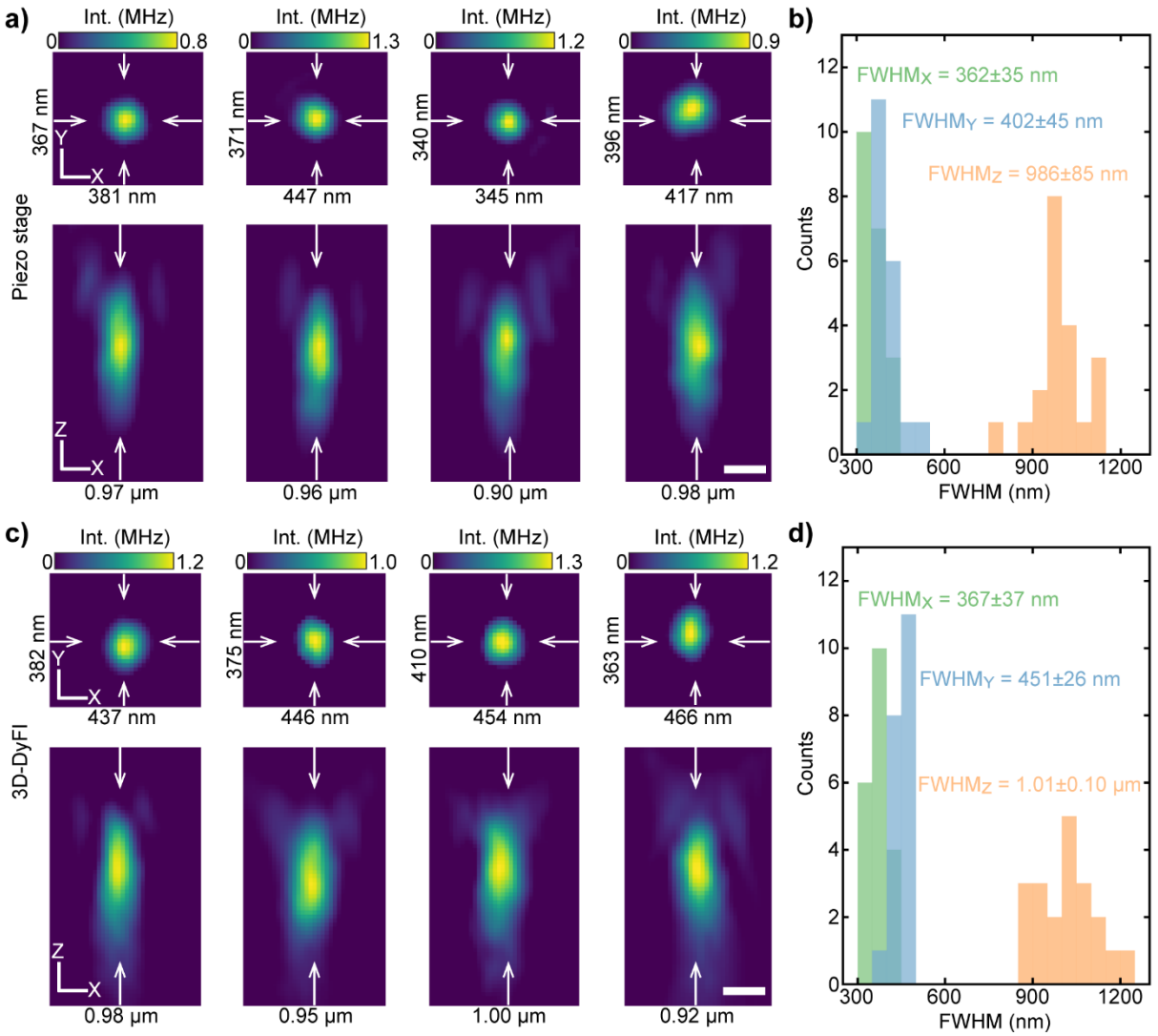


Figure S6. The resolution characterization with 50 nm red fluorescent spheres in the center of scanning range of dynamics focusing lens. Images of four typical 50 nm red fluorescent microspheres (635 nm excitation) in the x-y and x-z planes, obtained using the Piezo stage **(a-b)** and 3D-DyFI **(c-d)** systems. The upper panel shows the PSF in the x-y plane, and the lower panel shows the PSF in the x-z plane. Pixel size: 50 nm. Scale bar: 500 nm. **(b, d)** Statistical histogram of the spatial resolution distribution in the x, y, and z axis for the piezo stage-based confocal microscopy **(b)** and 3D-DyFI **(d)** systems. n = 20. The color indicates fluorescence intensity in (**a**) and (**c**).

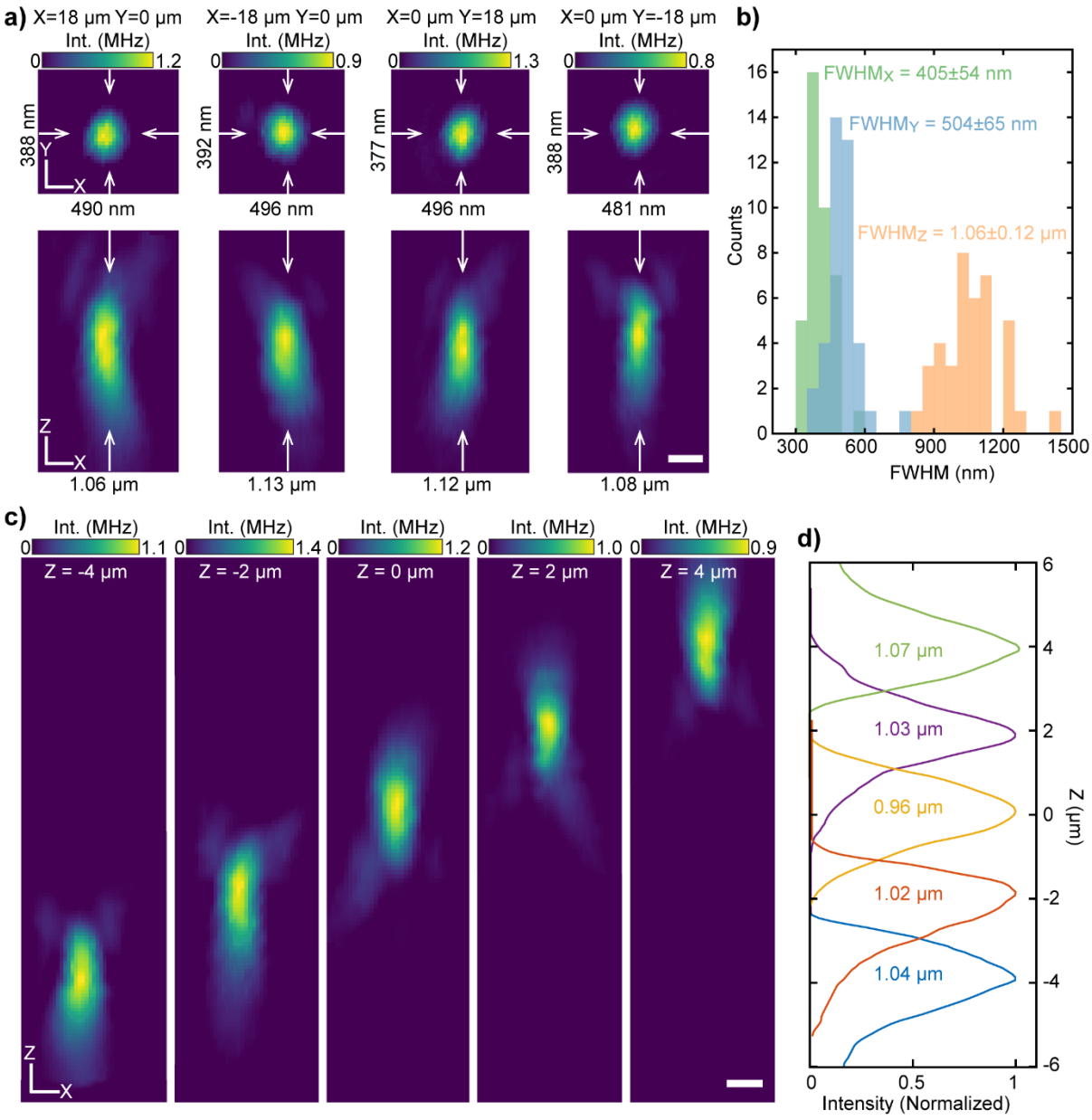


Figure S7. 3D spatial resolution examination in various positions in 3D with 50 nm red fluorescent beads. (a) Images of four typical 50 nm fluorescent microspheres located at the boundary of the maximum scanning range, with positions at (18, 0), (-18, 0), (0, 18), and (0, -18) in the x-y plane. The upper panel shows the PSF in x-y plane, while the lower panel shows the PSF in x-z plane. Scale bar = 500 nm. **(b)** The statistical histogram of spatial resolution distribution in x, y, and z directions, which shows mean values of 405 ± 54 nm, 504 ± 65 nm, and 1.06 ± 0.12 μm in x, y, and z. n=39. **(c)** Fluorescent microsphere images at different z-depths (-4 μm, -2 μm, 0 μm, 2 μm, and 4 μm). Scale bar = 500 nm. **(d)** Plot of the

intensity distribution of the PSFs from **(c)** with the fitted FWHM values to determine the corresponding axial resolution.

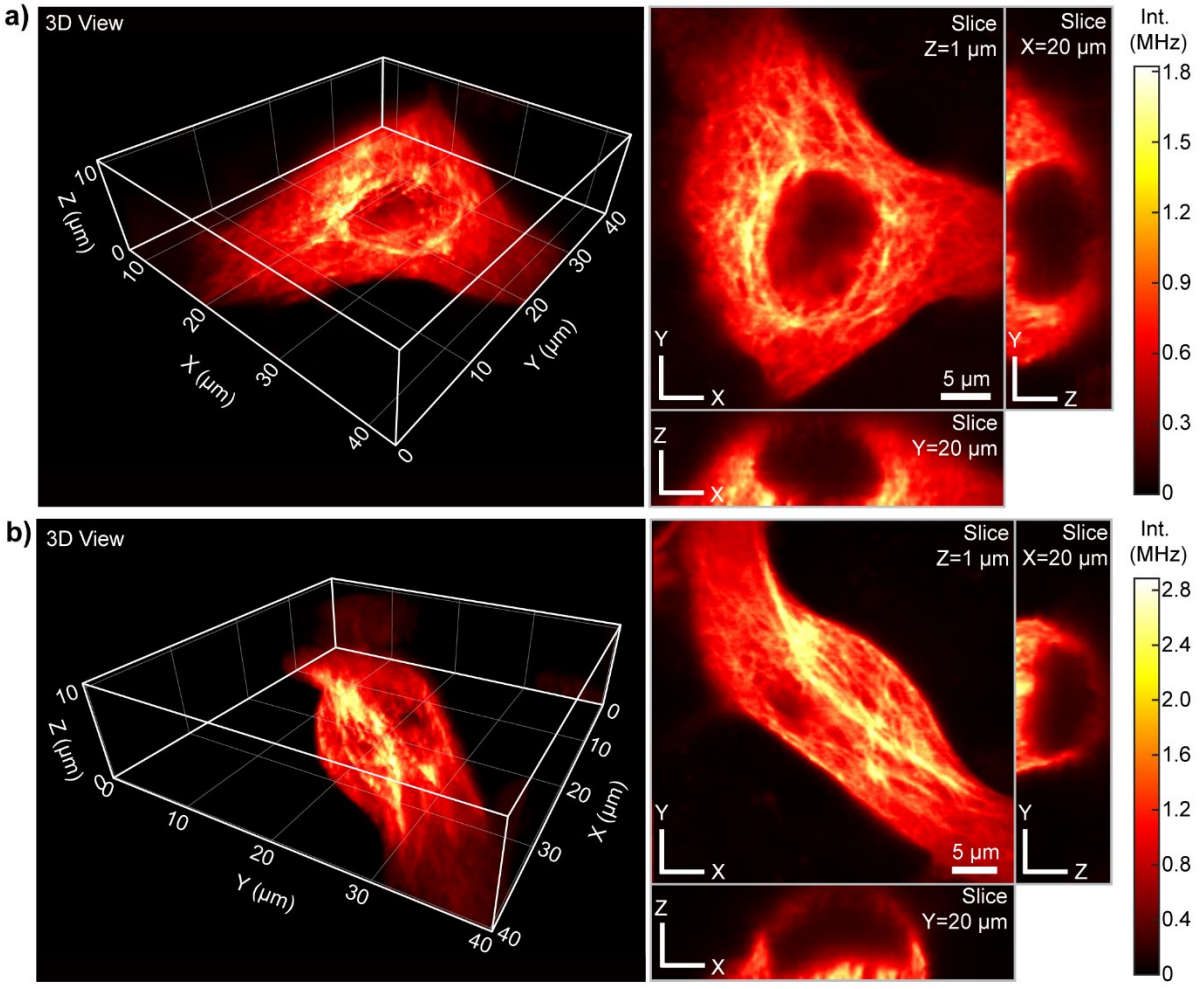


Figure S8. 3D imaging of microtubules of live Hela cell. The microtubules were labeled with mBaojin fluorescent protein. The left panel shows 3D live cell imaging. The right panel shows the section image in the x-y, x-z, and y-z planes. The sectioning positions are labeled in the upper right corner of the images. The color indicates fluorescence intensity.

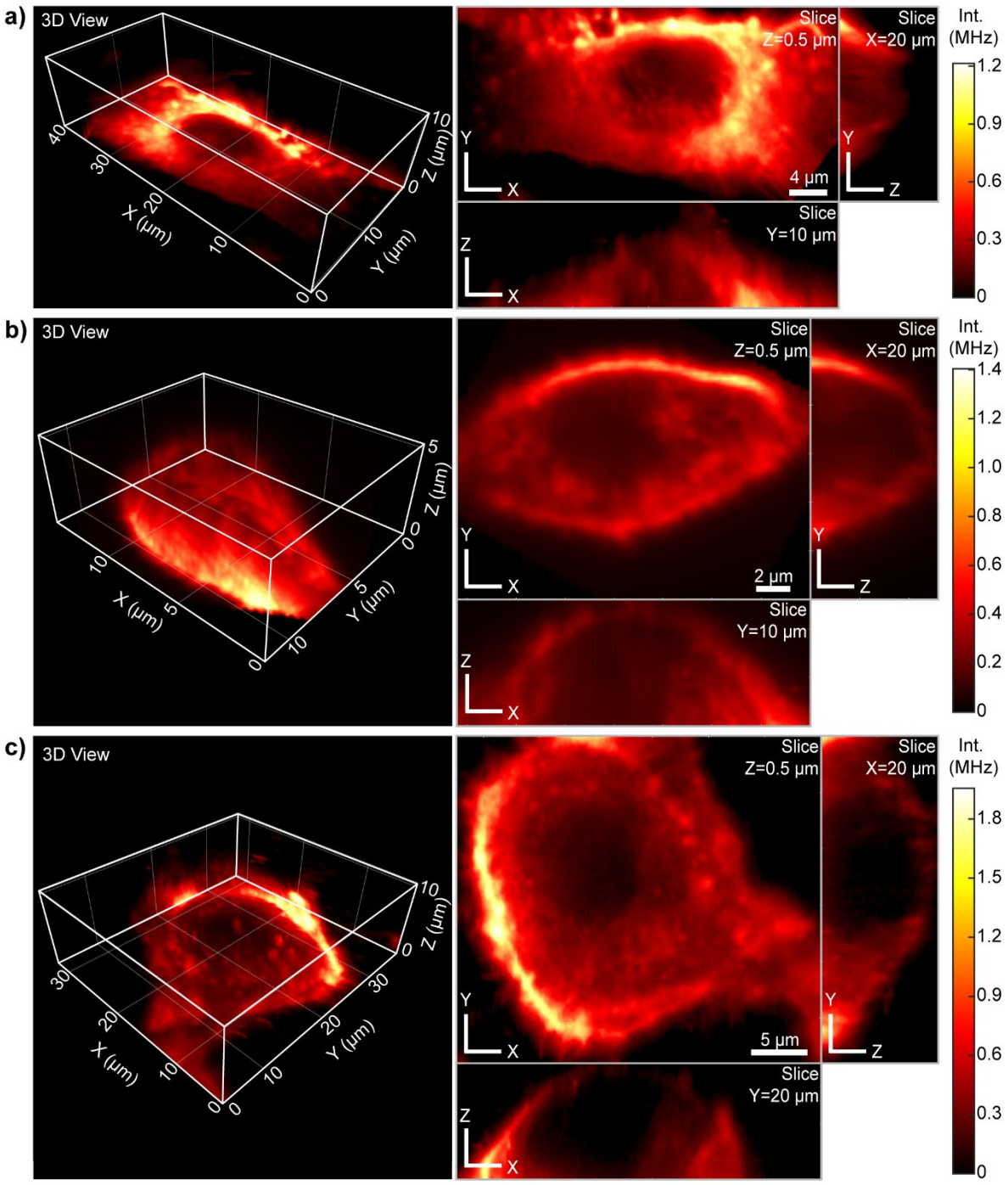


Figure S9. Three-dimensional imaging of DiD-stained Hela cell membranes with 3D-DyFI (a-c). The left panel shows 3D live cell imaging. The right panel shows the section image in the x-y, x-z, and y-z planes. The sectioning positions are labeled in the upper right corner of the images. The color indicates fluorescence intensity.

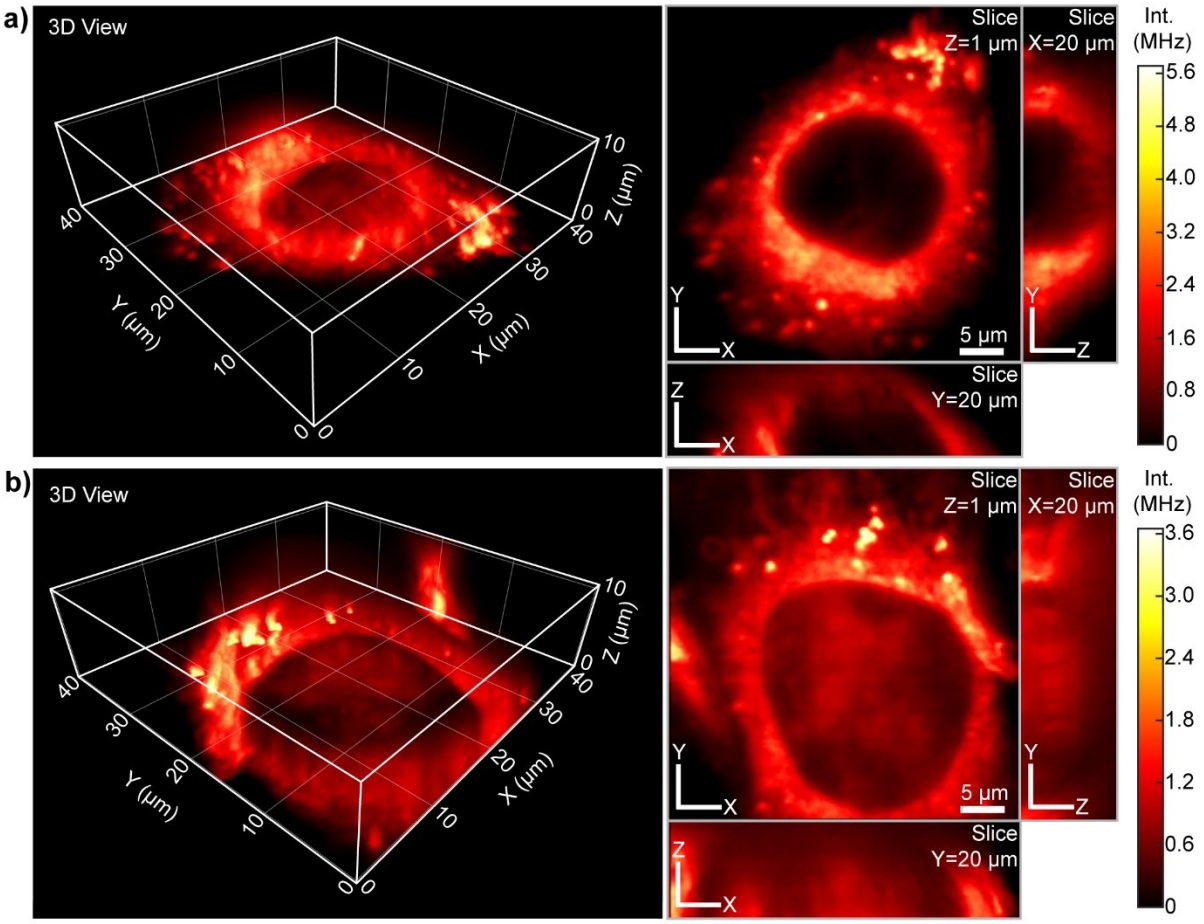


Figure S10. 3D imaging of Nile Red labeled live L929 cell. The lipid droplets (bright spots) and cell membrane were labeled by Nile Red. The left panel shows 3D live cell imaging. The right panel shows the section image in the x-y, x-z, and y-z planes. The sectioning positions are labeled in the upper right corner of the images. The color indicates fluorescence intensity.

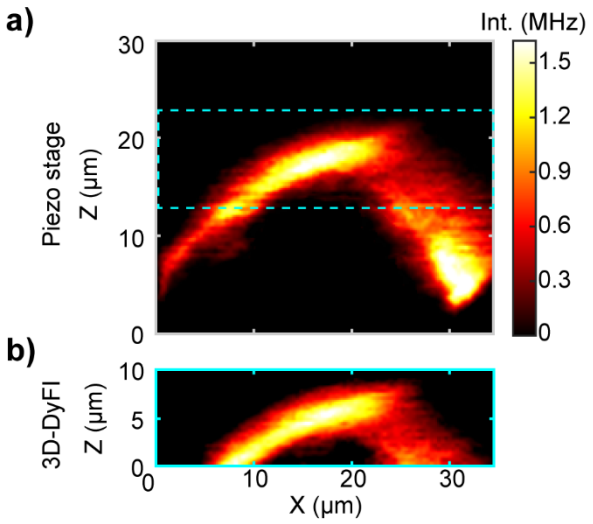


Figure S11. Thick cell imaging with 3D-DyFI. **(a)** x-z plane image of DiD-stained Hela cell membrane obtained with piezo stage-based confocal microscopy. **(b)** x-z plane image of DiD-stained Hela cell membrane obtained with 3D-DyFI. The imaging area is selected to the top part of cell as marked with dash box. Pixel size in x: 0.1 μm; Pixel size in z: 0.2 μm. The color indicates fluorescence intensity.

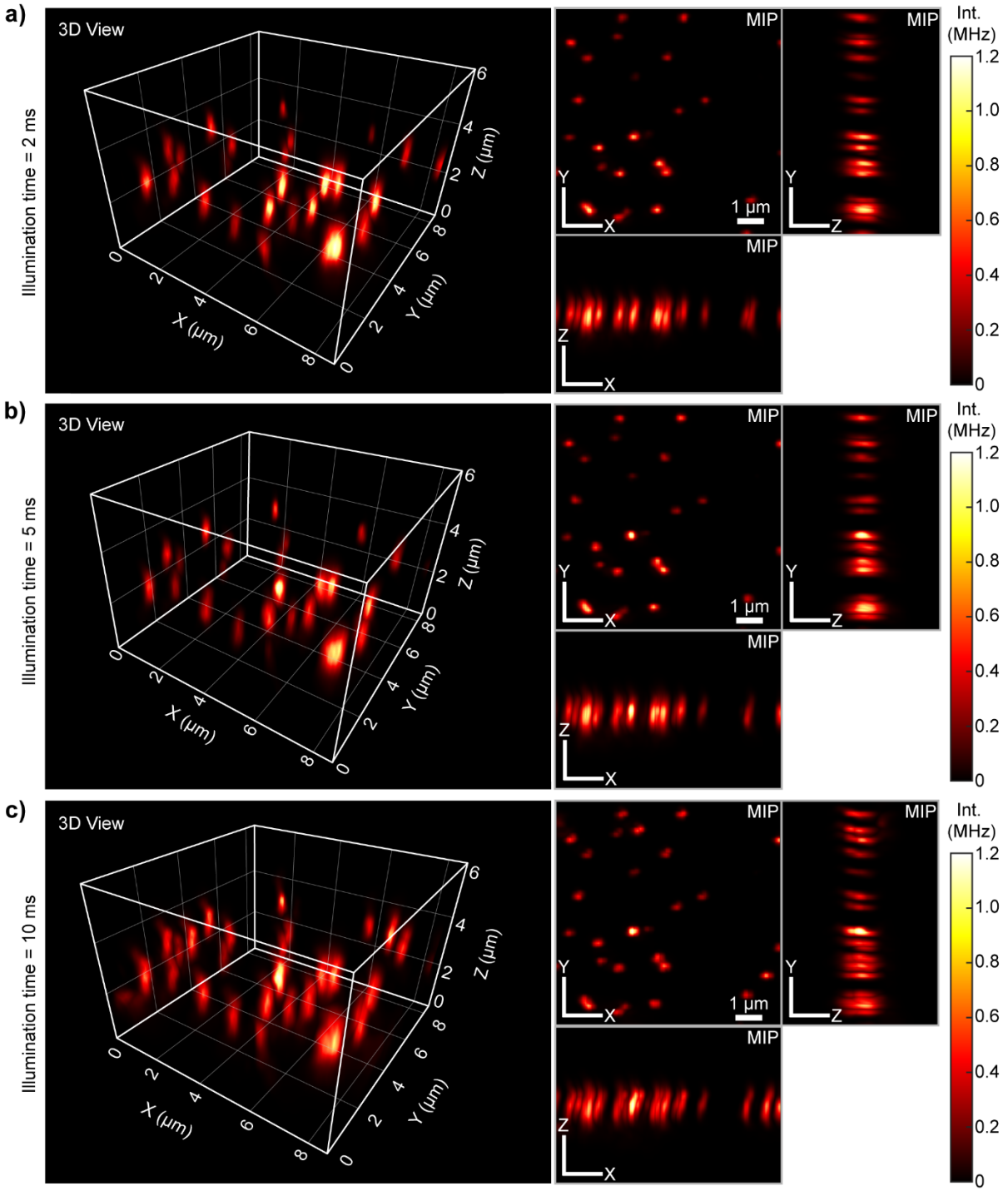


Figure S12. 3D imaging of 50 nm red fluorescent microspheres with different pixel illumination times. Microspheres located at coverslip were imaged with 3D-DyFI with illumination times of 2 ms **(a)**, 5 ms **(b)**, and 10 ms **(c)**. The left panels show 3D views, while the right panels display maximum intensity projection (MIP) images along the x-y, y-z, and x-z planes. Scale bar: 1 μm. The color indicates fluorescence intensity.

Supplementary Note 1. Specifications of the galvo mirrors and motors in remote focusing system.

The mirrors are dielectric coated, with Ravg≥98% @ 405 - 700 nm at angle of incidence from 30 to 60 degrees. The size of the mirror is 30 mm × 35 mm. The linearity of motors is 99.9%. The long-term drift of motors over 8 hours: < 0.5 mRad. The machinery scan angle of XY scanning is ± 11°. The control voltage range is ± 5 V. The repeatability, indicated by relative root mean square error (RRMSE), are 0.058, 0.064, and 0.033 for x, y, and z, respectively (Figure S2 and S3).

Supplementary Note 2.

Minimize aberration with scanning tube lens

The excitation field distortion is a common problem for 3D dynamic focusing system[1]. In piezoelectric stage-based imaging system, the sample is moved step by step. Therefore, in this situation the excitation and detection path remain stationary, which avoided the aberration and field distortion. However, in a 3D dynamic focusing system, the galvanometer mirror is tilted to steer the beam, and the light will be deflected off its initial straight (optical axis) path, which could induce severe aberration. Instead of using standard spherical or achromatic doublet lenses for relaying the beam after the 3D dynamic focusing system, we employed a laser scanning tube lens. This laser scanning tube lens can create a telecentric system, providing flat field and aberration reduced focused laser after objective lens[1].

To demonstrate the advantage of using a laser scanning tube lens over conventional achromatic doublet lenses in 3D-DyFI, we conduct 3D imaging of 200 nm red fluorescent microspheres under both conditions (Fig. **S13**). In 3D-DyFI with achromatic doublet lenses, the image of fluorescent microspheres exhibits severe distortions and spherical divergence, especially in the area further away from the center of the image (Fig. **S13a**). In contrast, the laser scanning tube lens delivers uniformly clear images across the entire imaging field (**Fig. S13b**), showing pronounced aberration-correcting capabilities. In this microsphere sample, microspheres are located on the surface of the coverslip. Theoretically, these microspheres should have the same axial position, as shown in the image taken with laser scanning tube lens base 3D-DyFI (**Fig. S13b**, maximum projected x-z and y-z images). However, the image taken with the achromatic doublet lens base 3D-DyFI shows obvious axial position variation (**Fig. S13a**, maximum projected x-z and y-z images). These results further validate the correction capacity of the laser scanning tube lens in 3D-DyFI, which ensures a flat and uniform field scanning. Additionally, the use of laser scanning tube lenses significantly enhances the axial resolution of 3D-DyFI. As shown in **Fig. S13**, it is evident that the using of conventional achromatic doublet lenses produced an axial PSF elongation by approximately 2 times compared with that use scanning tube lens. Altogether, the using scanning tube lenses in 3D-DyFI provides distortion and aberration reduced excitation,

ensuring accurate and high-resolution 3D imaging. The aberration can be further reduced by using adaptive optics or spatial light modulators, at the cost of system complexity [2-4]. Another way to reduce the aberration is use of long working distance objective lenses [5].

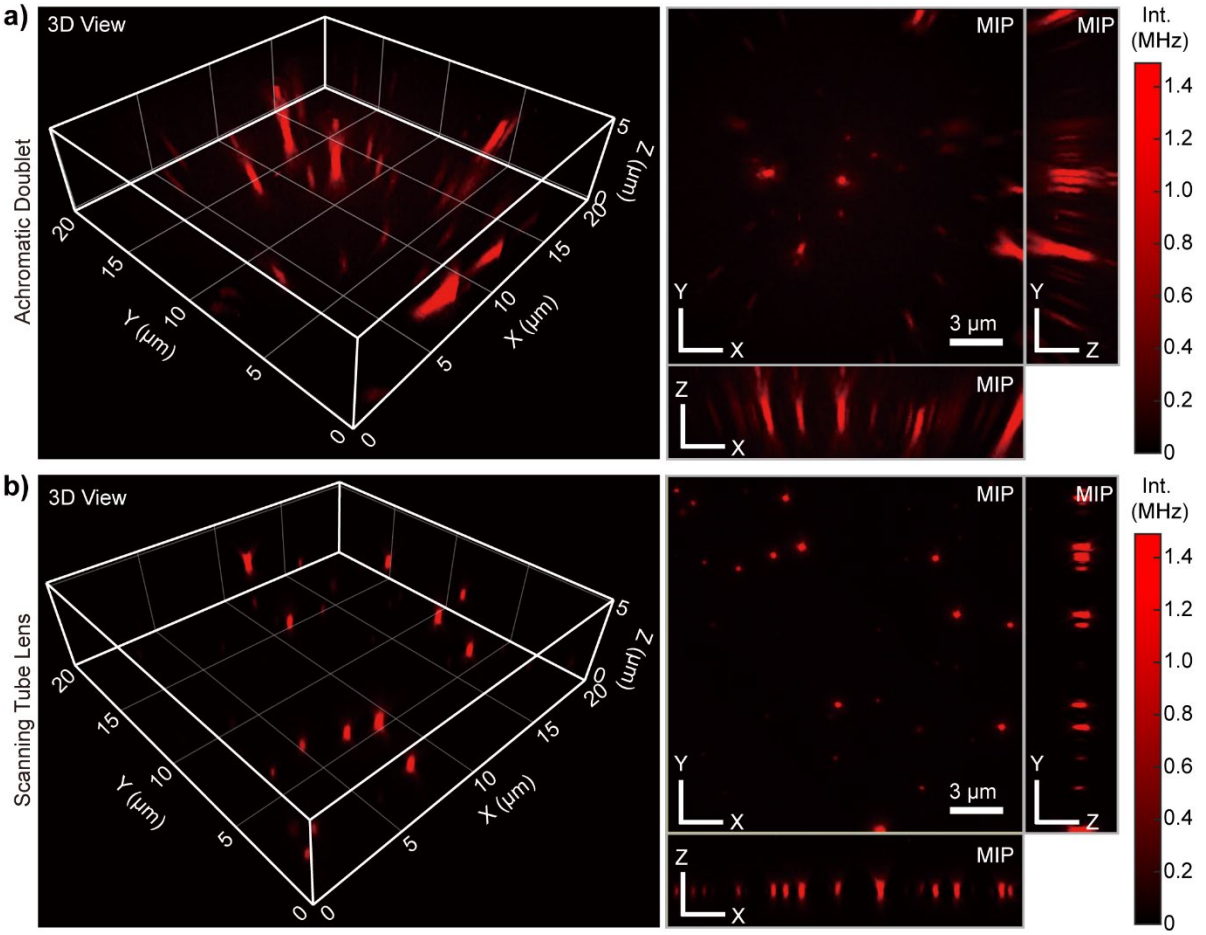


Figure S13. Aberration correction comparison in 3D-DyFI: achromatic doublet lens vs. scanning tube lens. Three-dimensional imaging of 200 nm fluorescent beads on a glass slide in PBS buffer, using an achromatic doublet lens **(a)** and a scanning tube lens **(b)**, with corresponding maximum intensity projection in the x-y, x-z, and y-z planes shown in the right panel. The color indicates fluorescence intensity.

Supplementary References

1 Negrean, A. & Mansvelder, H. D. Optimal lens design and use in laser-scanning microscopy. *Biomed. Opt. Express* **5**, 1588-1609 (2014).

2 Yang, J.-T. *et al.* Remote focusing with dynamic aberration elimination by model-based adaptive optics. *Optics & Laser Technology* **169**, 110126 (2024).

3 Yang, Y., Chen, W., Fan, J. L. & Ji, N. Adaptive optics enables aberration-free single-objective remote focusing for two-photon fluorescence microscopy. *Biomed. Opt. Express* **12**, 354-366 (2021).

4 Jiang, H. *et al.* Long-range remote focusing by image-plane aberration correction. *Optics Express* **28**, 34008-34014 (2020).

5 Qi, Y. *et al.* Remote-focusing microscopy with long working distance objective lenses. *Appl. Opt.* **53**, 3473-3478 (2014).